

Learn Prolog Now!

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5.5 Exercises

Exercise 5.1 How does Prolog respond to the following queries?

1. $X = 3 * 4$.
2. X is $3 * 4$.
3. 4 is X .
4. $X = Y$.
5. 3 is $1 + 2$.
6. 3 is $+(1, 2)$.
7. 3 is $X + 2$.
8. X is $1 + 2$.
9. $1 + 2$ is $1 + 2$.
10. $\text{is}(X, +(1, 2))$.
11. $3 + 2 = +(3, 2)$.
12. $*(7, 5) = 7 * 5$.
13. $*(7, +(3, 2)) = 7 * (3 + 2)$.
14. $*(7, (3 + 2)) = 7 * (3 + 2)$.
15. $7 * 3 + 2 = *(7, +(3, 2))$.
16. $*(7, (3 + 2)) = 7 * (+(3, 2))$.

Exercise 5.2

1. Define a 2-place predicate increment that holds only when its second argument is an integer one larger than its first argument. For example, $\text{increment}(4, 5)$ should hold, but $\text{increment}(4, 6)$ should not.
2. Define a 3-place predicate sum that holds only when its third argument is the sum of the first two arguments. For example, $\text{sum}(4, 5, 9)$ should hold, but $\text{sum}(4, 6, 12)$ should not.

Exercise 5.3 Write a predicate addone/2 whose first argument is a list of integers, and whose second argument is the list of integers obtained by adding 1 to each integer in the first list. For example, the query

?- addone([1, 2, 7, 2], X).

should give

$X = [2, 3, 8, 3]$.

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